



CEU San Pablo  
School of Engineering  
Software Engineering

# Trastervere

A project submitted of the requirements for the Software Engineering subject in  
Bachelor's of Data Science and Engineering

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# 1 Project Vision and Concept

The Trastevere Web-App project, conceptually defined during Weeks 9 and 10, was conceived as a digital platform centered around the creation, sharing, and discovery of culinary recipes. The application connects users who wish to upload their own creations with a community eager to explore new culinary ideas. The central idea revolves around social interaction and premium content monetization. By defining a double-tier subscription model (Basic and premium) early in Week 12, Trastevere aims to provide open access to casual users while offering exclusive functionalities (such as unlimited access or personalized experiences) to its premium users. The application facilitates the storage and sharing of recipes, user interactions through ratings and comments, and the structured organization of the content.

## 2 Development Approach and Timeline Execution

The project's development was originally managed following a traditional iterative and incremental methodology, supported by Gantt charts in early weeks for estimating effort, dependencies, and tracking workloads in parallel. However, as the complexity of the system grew, the team faced a critical delivery on Week 18 (for version 2) and finally adopted an Extreme Programming (XP) approach. This transition to XP prioritized continuous communication, adaptability, and code quality over rigid early-stage planning. The principal XP practices incorporated included:

- **Pair Programming:** Used heavily in Week 18 for tasks as complex as integrating the external Redsys Virtual POS and for system testing.
- **Continuous Integration and Small Releases:** The team focused on incremental and functional deliveries (v1, v2, v3), deploying immediate fixes for functionalities like the password recovery flow and UI alignments during Week 19.
- **Collective Ownership and Refactoring:** Code ownership was shared across the team to resolve IT tickets effectively in Week 19, and major refactoring was undertaken to clean up the codebase after the v2 delivery to transition the database to the v3 schema

## 2.1 Development Phases

- **Initiation and System Design** (*Weeks 9–11*): Brainstorming, initial wireframes, local PostgreSQL database implementation, and the critical selection of the Python-based Flet framework for frontend and backend integration.
- **Major Functionalities – v1** (*Weeks 12–16*): Implementation of initial login, home feed, publication upload, and search capabilities. This phase concluded with cross-beta testing procedures – delivering the beta in Week 14 and testing the "El Volcán" group app in Week 15 – leading to the official v1 submission in Week 16.
- **Premium Features – v2** (*Weeks 17–18*): Execution of database migrations ( $v1 \rightarrow v2$ ). The focus went from basic functionalities to monetization and social interactions, including the follow system, ratings, weekly coupons, and the Redsys payment gateway integration. v2 was submitted in Week 18.
- **Validation, Migration, Final Release – v3** (*Weeks 19–21*): Execution of the final database migration ( $v2 \rightarrow v3$ ). Implementation of comment systems, secure password recovery, advanced JSON-based search history, and bug ticket resolutions. The final project submission and delivery occurred in Week 21.

## 3 System Features and Capabilities

Trastevere Web-App provides functionality for both content creators and consumers, incorporating advanced search capabilities and integrated social features.

### 3.1 Authentication System and User Profiles

The authentication and profile systems that manage essential personal data were first integrated into the application during Week 12. Security was heavily prioritized in the later stages; in Week 19, the team implemented a secure password recovery system utilizing the Resend email client to send secure recovery tokens to users. Additionally, in Week 19, an "About" section that allows public visitors to view software information without needing to log in was introduced.

## 3.2 Major Functionalities (Feed, Drafts, Tables)

The center of the application features a home feed where users can explore the recipes published. To ensure data persistence and a seamless UX, the drafts system (temporal savings) was developed during Week 19, allowing users to save intermediate recipe statuses in a dedicated "Table" to prevent loss of unsaved process. This feature comes from a ticket received during the testing phase of *v2*

## 3.3 Subscription Model

A monetization system was designed to distinguish between basic and premium users. The most critical update occurred in Week 18, where an external Virtual POS (Redsys) via a secure tunnel (SHA--256 and 3DES) was integrated to handle transactions securely without storing raw credit card data.

- **Basic Plan:** Users are limited to viewing 4 recipes per day and are restricted to a single use of the platform's *"I'm feeling lucky"* feature.
- **Premium Plan:** Offers limitless daily recipes and infinite usage of the *"I'm feeling lucky"* button. Premium unlocks unique UI features designed in Weeks 17 and 18, such as the "Recipe of the Week" logic, exclusive weekly coupons, and visual themes for the application only for subscribers.

## 3.4 Social Features and interaction Systems

To gather a community around culinary publications, updates throughout Weeks 17 to 19 introduced the interaction tools between users:

- **Follow System:** A user follow/unfollow interface to keep track of favorite creators, implemented across Weeks 17 and 18.
- **Ratings and Comments:** A 1-to-5 star system for recipes (developed starting Week 17), paired with an asynchronously loaded comments section built in Week 19.
- **Search History:** A mechanism that tracks and stores the user's search queries, successfully transitioned in Week 19 using JSON persistence fields in the *v3* database.

## 4 Architecture and Data Management

The platform operates utilizing a local client-server architecture defined early in Week 10, relying on Python as backend technology and PostgreSQL for data persistence.

- **Frontend Framework:** Trastevere implemented Flet (a Python-based Flutter framework) selected in Week 10 to natively manage responsive grids, card-based layouts, and modern UI components.
- **Database and Migrations:** The relational PostgreSQL database handles tables for users, publications, and collections. To mitigate problems during migrations, the whole database schema was built using SQLAlchemy. The development team executed a strict migration strategy via version control, handling the migrations using advanced data formats, such as JSONB, which was used for the ingredient tagging system (Week 10) and later used extensively for search history storage (Week 19)

## 5 User Experience and Interface Design

The Trastevere Web-App design philosophy aimed to enhance the user experience continuously, specifically driven by feedback and ticket resolutions:

- **Internationalization:** Interface translation and the option of displaying the app in English/Spanish were implemented during Week 18.
- **Customization and UI:** A theme color customization feature was scheduled and integrated in Week 18 so users (premium) can adapt the app's visual style.
- **Error Prevention and Control:** Strict usability updates were applied in response to user testing during Week 19, such as adding a reliable "Cancel" option in the profile editing menu to prevent unintended data overwrites.

## 6 Testing Protocols and Developer Guidelines

For testing and assessing the Trastevere application, the following procedures reflect the platform’s lifecycle standards:

- (i) **Cross-Beta Testing:** Early versions of the application were rigorously evaluated. The Trastevere team delivered their beta to the ”El Volcán” group and reciprocally tested the ”El Volcán” app.
- (ii) **Issue Resolution Logging:** Over the final development phase in Weeks 19-21, regression testing was paramount. 11 new tests (e.g., T\_LOGIN\_04, T\_USER\_05, T\_PUB\_05-07, T\_ABOUT\_05-07) were verified specifically to ensure that previously solved IT tickets—like draft persistence (T\_INT\_08) and database recovery tokens (T\_DB\_05)—functioned correctly
- (iii) **Database Setup:** When initiating a new local environment, ensure the connections between the Python models and the PostgreSQL tables are functional (originally verified through testing designed in Week 10), and that the v3 schema migrations created in Week 19 are properly executed.

## 7 Technical Stack and INtegrations

The technological foundation of the Trastevere Web-App relies on tools sequentially incorporated throughout the semester:

- **Database Management:** PostgreSQL defined in Week 11, featuring strict schema versioning and JSONB support.
- **Backend:** Flet Framework (Python-based Flutter) selected in Week 10 for responsiveness.
- **Third-Party Integrations:** Redsys Virtual POS integrated in Week 18 for secure financial transactions via Flask, and Resend APIs implemented in Week 19 for secure email client communication and password resets.